

Valter Nunez

Software Engineer — AI / LLM Systems | MSc Data Science, LSE (London)

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Summary

Backend engineer who builds **LLM-integrated products**, with an **agent-orchestrated development** workflow that ships at small-team scope solo. Five years of distributed-systems work, plus a quantitative foundation and Bayesian-inference projects. Now studying **MSc Data Science at LSE** in London (UK Graduate visa, 18 months, no sponsorship required); seeking Skilled Worker sponsorship to stay long-term.

AI / LLM Engineering

- Integrated an LLM (**Google Gemini**) into a production SaaS, an "Improve with AI" assist with the controls a real feature needs: **usage metering**, daily request/token **limits**, alert thresholds, a **fallback model**, graceful degradation, and **schema-enforced structured output** (Nexus).
- Designed an **orchestrated multi-agent development workflow** to build Nexus solo: specialist Claude Code agents (implement, review, and nine read-only auditors) running in parallel across isolated **git worktrees**, handing off via API contracts and diffs; human-in-the-loop, ~1,200 lines of command definitions.

Selected Projects

- Refract** — agent-first responsive-screenshot tool: renders a URL at N viewports and returns images plus structured findings (overflow, clipped text, tap-target issues) as agent-actionable JSON; on npm as an **MCP server**, CLI, and Node library (TypeScript, Playwright); MIT.
- Ballon d'Or attention study** — Bayesian two-stage decomposition (PyMC, ~210 player-seasons) isolating an unexplained media-attention residual; shipped as a D3 scrollytelling site + Quarto report.
- World Cup 2026 momentum** — natural-experiment data pipeline (local scrape → GitHub Actions → Pages) that regenerated daily.

Experience

Backend Engineer Sep 2021 – Mar 2026
ZeBrands — high-traffic e-commerce *Remote, Mexico*

- Co-architected and built **OZ**, an event-driven platform that sustains sales through ERP outages, decoupling 83% of revenue-critical sales flows from synchronous ERP calls (FastAPI, PostgreSQL, RabbitMQ, Redis); now live across ~195 stores (Mexico, expanding to Brazil), sustaining ~MX\$102M over the nine-day Hot Sale peak.
- Co-designed and built multi-carrier shipping middleware with automated least-cost routing (~78% lower shipping spend); maintained existing client-facing microservices and GraphQL data APIs, and stood up Grafana observability.

Co-Founder & Developer 2017 – 2019
Laiter — software studio *Mexico*

- Built Django web apps and the early usage-analytics dashboards clients ran decisions on.

Education

MSc Data Science 2026 – 2027
London School of Economics (LSE) *In progress, London*

B.S. Computer Science & Technology 2016 – 2021
Tecnológico de Monterrey (ITESM) *Avg 90/100*

Semester abroad at RMIT Melbourne (Avg 94/100). IBM Data Science Professional Certificate (in progress).

Technical Skills

AI / LLM: LLM integration (Gemini), usage metering & rate-limiting, structured output, agent-orchestrated development (Claude Code, git worktrees)

Data Science: Python, pandas, NumPy, scikit-learn, PyMC, Bayesian inference, statistics

Backend: FastAPI, Django, GraphQL, Celery, PostgreSQL, Redis, RabbitMQ

Infra: Docker, GitHub Actions, Grafana; event-driven & microservice design

Languages spoken: Spanish (Native), English (C2), German (A2), Italian (A1)